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Project: Vampire Wars Combat Simulator
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Description:
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A command-line dice game that simulates a turn-based battle between a Human and a Vampire. This script demonstrates core programming fundamentals including custom functions, nested while loops, conditional logic, and random number generation for dynamic combat resolution. Players dictate the number of rounds, and health is calculated in real-time based on dice roll totals.
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import random
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# Roll the given number of dice and return the results
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def roll_dice(number_of_dice):
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    dice_rolls = []
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    for i in range(number_of_dice):
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        dice_rolls.append(random.randint(1, 6))
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    return dice_rolls
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```
# Calculate the total value of all dice
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def calculate_total(dice_rolls):
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    total = 0
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```
    for dice in dice_rolls:
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```
        total += dice
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```
    return total
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```
# Start the game
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play_game = input("Would you like to play Vampire Game? [y/n]: ")
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while play_game == "y":
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    # Starting health
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    human_health = 100
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    vampire_health = 100
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    # Get number of rounds
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    rounds = int(input("Enter number of rounds 1-5: "))
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    while rounds < 1 or rounds > 5:
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```
        rounds = int(input("Please enter rounds between 1-5: "))
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    current_round = 0
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# Play rounds until finished or someone dies
while current_round < rounds and human_health > 0 and vampire_health > 0:
    current_round += 1

    print(f"\n--- Round {current_round} ---")

    # Roll dice for both players
    human_roll = roll_dice(5)
    vampire_roll = roll_dice(5)

    # Calculate roll totals
    human_total = calculate_total(human_roll)
    vampire_total = calculate_total(vampire_roll)

    print("Human rolled:", human_roll)
    print("Human total:", human_total)

    print("Vampire rolled:", vampire_roll)
    print("Vampire total:", vampire_total)

    # Total roll = damage dealt
    human_damage = human_total
    vampire_damage = vampire_total

    # Apply damage
    vampire_health -= human_damage
    human_health -= vampire_damage

    print(f"Human dealt {human_damage} damage")
    print(f"Vampire dealt {vampire_damage} damage")

    print(f"Human health: {human_health}")
    print(f"Vampire health: {vampire_health}")

# Show battle result
print("\n--- Battle Result ---")

if human_health <= 0 and vampire_health <= 0:
    print("Draw! Both died.")
elif human_health <= 0:
    print("Vampire wins!")
elif vampire_health <= 0:
    print("Human wins!")
elif human_health > vampire_health:
    print("Human wins by higher health!")
elif vampire_health > human_health:
    print("Vampire wins by higher health!")
else:
    print("Draw!")

# Ask to play again

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```
play_game = input("\nPlay again? [y/n]: ")  
print("Thanks for playing!")
```